Excel Sales Dashboard

Project Overview

This project involved designing a sales dashboard in Excel to analyze sales based on product categories. The dataset provided contained variables such as Order ID, Order Date, Aging, Product Category, Sales, Quantity, Profit, Region, and Months. The goal was to add user control for product categories, allowing customers to choose a category and view the trend month-by-month and product-by-product, ultimately creating a sales dashboard that breaks down sales by product category.

Steps Taken

Data Preparation

- Used the saved sample e-commerce database as the primary source for analysis.
- Accessed the Data Analysis tool under the Data tab and selected the Histogram function to analyze data distribution.
- Configured the histogram dialog box by assigning the raw data column to the Input reference box and defining the grouping criteria in the bin range reference box.
- Generated a binning table and histogram chart on the working sheet by specifying the output section and selecting the appropriate output check boxes.

Analysis Tables

- Prepared summary tables in a working sheet to analyze sales and profit on a month-wise basis, alongside a separate table for region-wise sales.
- Utilized the SUMIFS formula to calculate the overall total sales, quantity, and profit. This was done by passing the relevant numerical columns as the sum range and applying the product category column as the primary criteria.
- Populated the month-wise table by expanding the SUMIFS formula to evaluate multiple conditions, passing both the product category and the specific month columns as criteria arguments.
- Applied the exact same SUMIFS structure to generate the region-wise table, this time passing the region column as the secondary criteria to intersect with the selected product category.

User Control Integration

- Created a user control combo box for the product category.
- Inserted the combo box for the product category list in the dashboard sheet.

- Passed the input range "Working!Q2:Q5" and cell link "Working!R2" from the working sheet.
- Wrote the translation function by passing the first argument as Cell "Q1", selecting cell "R2" in the second argument, and typing zero in the third argument.

Visualization

- Created distinct column charts to visually represent the calculated data from both the month-wise and region-wise summary tables.
- Generated these visual insights by selecting the complete data ranges for each respective table, navigating to the insert tab on the ribbon, and choosing the Insert column chart option located under the Charts Panel.

Dashboard Design

- Linked the working tables with the combo box.
- Created a dashboard to present the charts and user controls in a single view.

Key Features of the Dashboard

- **Dynamic User Control:** Empowers users to seamlessly interact with the data by selecting a specific product category via the integrated combo box, which instantly filters the underlying metrics.
- **Automated Visualizations:** Translates complex data into clear, accessible visual insights by displaying responsive column charts that update automatically for both month-wise and region-wise tables.
- **Comprehensive Performance Analysis:** Fulfills the core business objective by analyzing sales across various product categories, revealing key performance patterns and trends on a month-by-month and product-by-product basis.

Technical Challenges & Solutions

During the development of the dashboard, structuring the SUMIFS formulas to accurately calculate sales, quantity, and profit across multiple dimensions (month-wise and region-wise) required meticulous logical alignment. The challenge of passing multiple interconnected arguments was resolved by strictly defining the sum ranges and criteria ranges within the formulas. By ensuring that the primary criteria dynamically linked to the translation cell generated by the user control combo box, the tables successfully updated in real-time without absolute referencing errors.

Conclusion The aim of the project was successfully met by analyzing the sales based on various product categories. The final dashboard ensures that users can pick a product category and can see trends month-by-month and product-by-product.